

A compact distribution-function model for ω Centauri embedded in dark matter

Model proposal, literature basis and validation programme

oCen_dm project notes

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Abstract

We propose a distribution-function (DF) model to constrain the present-day mass distribution of ω Centauri, allowing an extended DM halo and concentrated stellar remnants alongside self-gravitating stars. The primary spherical candidate is a regularized exponential stellar action DF with an anisotropy transition, combined with prescribed dark density profiles. Its five stellar parameters describe mass, action scale, envelope shape, anisotropy amplitude and transition scale. A two-parameter halo sampled in $\rho_{\text{DM}}(20\text{ pc})$ and scale radius, plus two remnant parameters, gives nine physical coordinates before light normalization and observational nuisance parameters. A generalized lowered-isothermal stellar DF in the same combined potential is the required independent control. The original linear-action exponential is retained only as an analytic precursor. Matched-tracer checks, rotation and flattening tests are required before treating spherical fits as final DM constraints. This note specifies a model proposal and validation programme; the regularized family is not yet implemented or validated.

1 Motivation and the physical problem

1.1 The target: Omega Centauri

ω Centauri (NGC 5139) is one of the Milky Way’s most massive globular clusters. It contains chemically distinct stellar populations and exhibits both rotation and flattening (Zocchi et al., 2017). These properties make it a more complex target than a spherical cluster with one uniform stellar population. A proposed origin is the surviving nucleus of an accreted dwarf galaxy whose outer body was removed by Galactic tides (Bekki and Freeman, 2003). This scenario motivates testing for a surviving DM component; it does not establish that the present cluster retains one.

We model the present-day surviving cluster, rather than reconstructing its entire progenitor in the equilibrium fit. The observational inputs are its surface-brightness profile and stellar kinematics: the existing project dataset contains 89 dispersion measurements, comprising 21 radial and 21 tangential HST proper-motion bins, 29 MUSE line-of-sight bins, and nine radial and nine tangential Gaia proper-motion bins. These are profile measurements, not a count of individual stars. The initial calculation adopts $D = 5.43$ kpc. “Remnants” here means unseen stellar remnants, such as white dwarfs, neutron stars and stellar-mass black holes; their possible concentrated mass component is distinct from an extended DM halo.

1.2 Why use a stellar distribution function?

Our scientific question is whether the measured light and stellar motions of ω Centauri require an extended dark component, and which present-day DM densities, especially $\rho_{\text{DM}}(20\text{ pc})$, they allow. A compact stellar cluster embedded in DM can be dominated by stars near its centre while the halo contributes at larger radii. Concentrated stellar remnants and orbital anisotropy can also affect the measured dispersions. We need a model that compares these possibilities using photometry, line-of-sight velocities, and both proper-motion components together.

The Jeans fits provide useful baseline mass and anisotropy constraints. They describe velocity moments, however, and a solution of the moment equations need not correspond to a non-negative stellar phase-space distribution. The purpose of the DF approach is to impose this stronger stellar consistency condition: a positive equilibrium $f_\star$ generates the stellar density and kinematics in the same gravitational potential. We seek the simplest stellar family that passes independent recovery tests and reproduces the data at their precision. Several freely mixed stellar DFs may provide approximation power, but their necessity must be established before interpreting their individual masses as physical populations.

This motivation does not require a DF for every gravitating component. In our stationary, smooth-potential fit, the unobserved DM and remnant velocities do not enter the stellar likelihood at fixed dark density. The recommended starting point is therefore one positive stellar DF plus parametrized dark density profiles, with stellar self-gravity retained. Explicit dark DFs are a separate option for stronger equilibrium checks and live N-body initial conditions. Comparing these two formulations is the purpose of Section 3; neither is assumed in advance to give a definitive DM detection or upper limit.

1.3 Scope of the first model

We start with a spherical, time-independent, nonrotating equilibrium at the adopted distance and with no central black hole. A BH is a subsequent extension. The spherical, single-tracer description is a controlled baseline for the actual rotating, flattened, multiple-population object; it is not a claim that those observed features are absent. Their effects must be tested before interpreting residuals as evidence for DM or another mass component.

We use AGAMA for actions, velocity integrals and gravitational potentials (Vasiliev, 2019). Both formulations in Section 3 determine the stellar density from a positive DF and include all mass components in Poisson’s equation:

$$\rho_\star(\mathbf{x}) = \int f_\star[\mathbf{J}(\mathbf{x}, \mathbf{v}; \Phi_{\text{tot}})] d^3v, \quad (1)$$

$$\nabla^2 \Phi_{\text{ext}} = 4\pi G(\rho_\star + \rho_{\text{rem}} + \rho_{\text{DM}}), \quad \Phi_{\text{tot}} = \Phi_{\text{ext}} - \frac{GM_\bullet}{r}. \quad (2)$$

Here Φ_{ext} is the potential of the extended mass components, not an external Galactic field. The initial stage sets $M_\bullet = 0$. For the first stellar fit, ρ_{DM} and ρ_{rem} may be prescribed density profiles. Giving them positive DFs is a stronger modelling choice, not a prerequisite for evaluating the stellar likelihood.

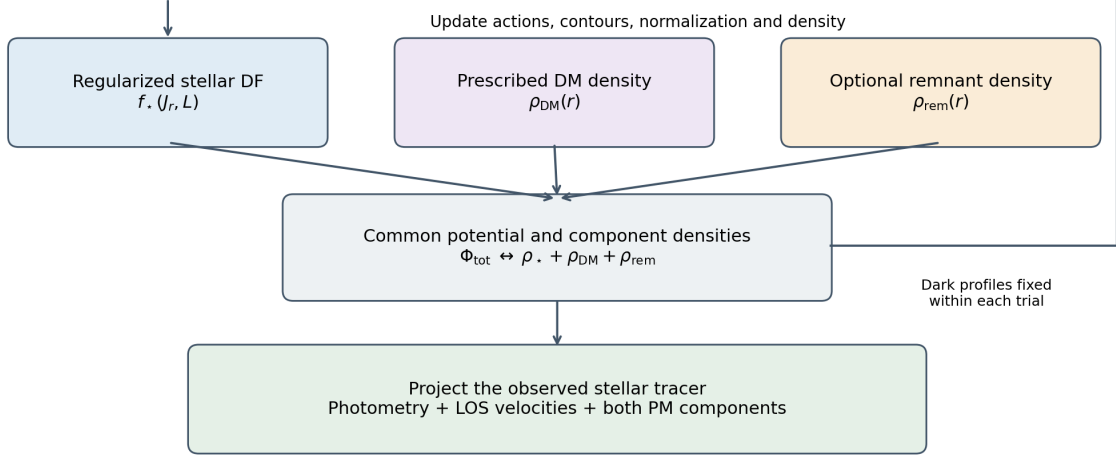


Figure 1: Recommended starting formulation: a positive stellar DF in the combined potential of stars and prescribed dark density profiles. At each trial, the stellar density and total potential are iterated together; the dark profiles remain fixed during that iteration. Their parameters vary between likelihood trials. Only the represented stellar tracer supplies the observables. This is a model diagram, not a simulation result.

2 A regularized exponential stellar DF

2.1 Actions and the required low-angular-momentum limit

In a spherical potential with $\Phi(\infty) = 0$, define the specific energy, angular momentum and radial action by

$$E = \frac{1}{2}v^2 + \Phi_{\text{tot}}(r), \quad L = |\mathbf{r} \times \mathbf{v}|, \quad (3)$$

$$J_r(E, L) = \frac{1}{\pi} \int_{r_{\text{peri}}}^{r_{\text{apo}}} \sqrt{2[E - \Phi_{\text{tot}}(r)] - L^2/r^2} dr. \quad (4)$$

Bound actions obey $J_r \geq 0$ and $L = J_z + |J_\phi|$. They have units pc km s^{-1} . The DF vanishes on unbound phase space.

For the smooth, finite-centre, no-BH baseline, regularity is an explicit condition. At fixed r and v_r , with $L = rv_t$, the radial-orbit limit gives

$$\lim_{v_t \rightarrow 0^+} \frac{\partial J_r}{\partial v_t} = -\frac{r}{2}, \quad \lim_{v_t \rightarrow 0^+} \frac{\partial f_\star}{\partial v_t} = r \lim_{L \rightarrow 0^+} \left(\frac{\partial f_\star}{\partial L} - \frac{1}{2} \frac{\partial f_\star}{\partial J_r} \right). \quad (5)$$

Consequently the nonzero DF derivatives must satisfy

$$\boxed{\lim_{L \rightarrow 0^+} \frac{\partial f_\star / \partial J_r}{\partial f_\star / \partial L} = 2.} \quad (6)$$

This is the spherical smoothness constraint discussed by Binney (2026). It concerns the local phase-space DF, not the probability density of the *magnitude* v_t , which includes a velocity volume factor. A positive steady DF can still fail this smoothness test; positivity, regularity and dynamical stability are separate requirements. The finite-potential limit also covers the proposed r^{-1} DM cusp, whose central potential is finite. A point mass or a divergent central potential requires a new limiting analysis: in a pure Kepler potential the derivative ratio is one. We will also check the joint action-space origin, which is not certified by a limit at fixed $J_r > 0$ alone.

The precursor $f = F(J_r + \eta L)$ has derivative ratio $1/\eta$. Only $\eta = 1/2$ satisfies equation (6); choosing that value does not make the whole system isotropic at every radius. We retain this family for analytic reproduction in Appendix B, not as the primary inference model. Its envelope and orbital coefficients both change the stellar density, so matching photometry cannot by itself determine its anisotropy.

2.2 Constant-DF contours for the reference model

Retain the exponential envelope of [Pascuale et al. \(2019\)](#), and adopt the spherical contour construction of [Binney \(2026\)](#). At any (J_r, L) , integrate

$$\frac{dL}{dJ_r} = -g(J_r, L), \quad g > 0, \quad (7)$$

towards decreasing J_r until reaching $(0, L_c)$. The endpoint label $L_c(J_r, L)$ defines the proposed DF:

$$f_\star(J_r, L) = A_\star \exp \left[- \left(\frac{L_c(J_r, L)}{J_{0,\star}} \right)^\alpha \right], \quad J_{0,\star} > 0, \quad \alpha > 0. \quad (8)$$

Set $f_\star(0, 0) = A_\star$ by continuity. Away from the origin, L_c labels a constant- f contour; it is not generally the angular momentum of a circular orbit at the *starting* orbit's energy. Along a contour, $df = 0$ implies $f_{J_r}/f_L = g$, wherever the ratio is defined. Thus building $g \rightarrow 2$ at small L enforces the required derivative ratio directly.

For an exactly ergodic control in a specified spherical potential, use

$$g_H(J_r, L) = \frac{\Omega_r(J_r, L)}{\Omega_t(J_r, L)}, \quad \Omega_r = \frac{\partial H}{\partial J_r}, \quad \Omega_t = \frac{\partial H}{\partial L}. \quad (9)$$

Then the contours conserve energy. An economical approximation suggested by the same construction is

$$g_H^{\text{app}} = \left[\frac{1}{2} + c \left(\frac{\Omega}{\kappa} - \frac{1}{2} \right) \right]^{-1}, \quad c = \frac{L}{L + J_r}, \quad (10)$$

where Ω and κ are the circular and radial epicycle frequencies. For a concrete pointwise implementation, evaluate their ratio on the circular orbit with angular momentum $\ell = L + 1.4J_r$, at the current point along the integration. This defines one vector field $g_H^{\text{app}}(J_r, L)$ for all contours. It is an approximation to test against equation (9), not a guarantee of exact isotropy. Positivity of g , smooth interpolation, unique endpoints and convergence of the contour solver must be verified.

Introduce anisotropy with

$$g(J_r, L) = g_H(J_r, L) \exp \left[-b(q) \sin \left(\frac{\pi c}{2} \right) \right], \quad q = J_r + L. \quad (11)$$

The choice of q is part of this proposal. Positive b favours radial orbits relative to the ergodic reference; negative b favours tangential orbits. For bounded b , the sine makes $g \rightarrow g_H \rightarrow 2$ at fixed $J_r > 0$ as $L \rightarrow 0$, regardless of anisotropy amplitude. With the approximate g_H , these biases are relative to a nearly isotropic reference and must be measured from the resulting moments.

The frequency map depends on the combined potential. Our reference prescription rebuilds that map, the contour labels and the numerical normalization as the trial potential changes. The converged model must satisfy both the DF prescription and Poisson's equation. This is a potential-dependent DF family, not an adiabatic evolution holding one function of actions fixed. A frozen reference-frequency map would define a different prescription and must be documented and tested separately.

2.3 An anisotropy transition from the outset

The primary candidate has

$$b(q) = b_{\text{out}} S(q/J_a), \quad S(x) = \frac{x^n}{1 + x^n}, \quad J_a > 0, \quad n = 2 \text{ fixed}. \quad (12)$$

The free stellar coordinates are $M_\star$, $J_{0,\star}$, α , b_{out} and J_a : five in total. This replaces the free constant η ; the net increase over the precursor is one parameter, not two. At $b_{\text{out}} = 0$, J_a is unidentifiable and should be removed from the explicitly isotropic comparison.

The choice is motivated by the isotropic centre and outward radial bias in the oMEGACat profiles (Häberle et al., 2025), together with the radial-to-tangential transition reported at larger radii in Gaia EDR3 (Vasiliev and Baumgardt, 2021). These samples probe different selections and radial ranges. Neither their agreement nor a required turnover in our own combined sample is assumed. If the matched-tracer data require a radial intermediate region and tangential outskirts, test

$$b(q) = b_1 S(q/J_1) + (b_2 - b_1) S(q/J_2), \quad 0 < J_1 < J_2, \quad (13)$$

with the same fixed n . This adds two coordinates relative to (b_{out}, J_a) . A distinct intermediate plateau requires sufficiently separated scales; merely imposing $J_2 > J_1$ does not ensure it.

The functions $b(q)$ are orbital-weight prescriptions, not $\beta(r)$. Orbits at one radius span many actions, and the potential itself changes across the fit. An approximately isotropic core and a radial-to-tangential profile must be demonstrated by the computed moments, not read off the signs of b_1, b_2 . Figure 2 illustrates the inputs only.

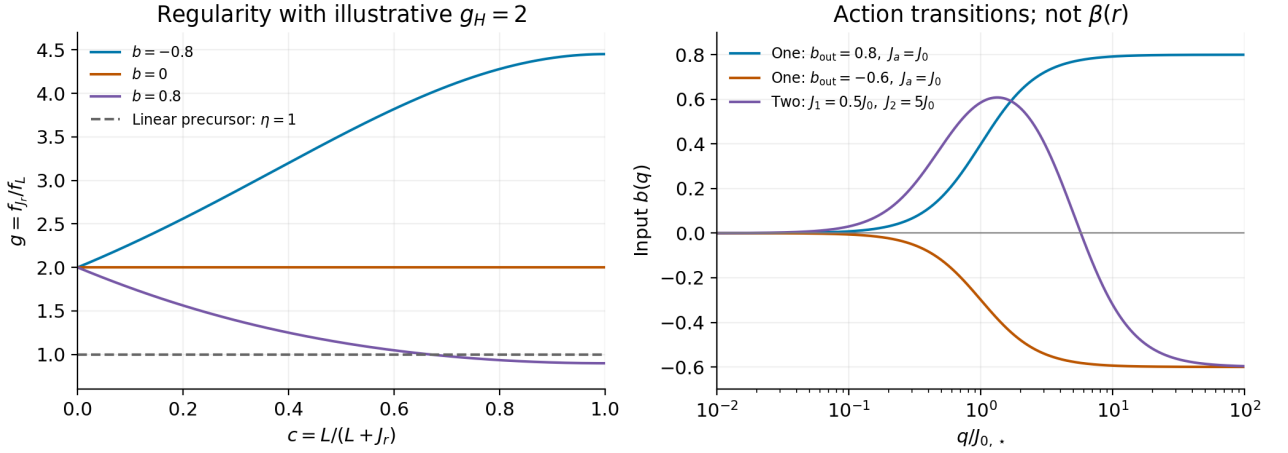


Figure 2: Dimensionless illustrations of the proposed orbital prescription. Left: for the illustrative harmonic reference $g_H = 2$, all bounded choices of constant b reach $g = 2$ at $c = 0$. The dashed line is the unregularized $\eta = 1$ precursor. Right: one- and two-transition input functions $b(q)$, with action units set by $J_{0,*}$ and $n = 2$. These are not computed $\beta(r)$ profiles, self-consistent equilibria or fits to data.

2.4 Numerical normalization, mass and light

The signed J_ϕ and positive J_z integrals reduce to $2L \, dL$. Normalize each trial with

$$A_\star = \frac{M_\star}{(2\pi)^3 \mathcal{I}_\star}, \quad \mathcal{I}_\star = 2 \int_0^\infty dJ_r \int_0^\infty L \, dL \exp \left[- \left(\frac{L_c(J_r, L)}{J_{0,\star}} \right)^\alpha \right]. \quad (14)$$

Use converged numerical quadrature, including the tail, rather than the precursor's analytic normalization. A harmonic-reference, $b = 0$ check gives $L_c = L + 2J_r$ and $\mathcal{I}_\star = J_{0,\star}^3 \Gamma(3/\alpha)/(2\alpha)$. This also checks the action-scale conversion to the $\eta = 1/2$ precursor.

Initially one effective stellar population has constant stellar mass-to-light ratio $\Upsilon_\star$, so $j_\star = \rho_\star/\Upsilon_\star$. $M_\star$ excludes any separately assigned remnant mass. The total M/L can still vary through DM and remnants. The velocity moments give

$$p_r = \int v_r^2 f_\star \, d^3v, \quad p_t = \frac{1}{2} \int (v_\theta^2 + v_\phi^2) f_\star \, d^3v, \quad \beta(r) = 1 - p_t/p_r. \quad (15)$$

The contour construction separates envelope and orbital controls more clearly, but changing $b(q)$ still changes the density. All stellar parameters must be fitted jointly to photometry and kinematics. It does not solve density–anisotropy covariance by construction. The one-tracer approximation requires the checks in Section 4.1 before a DM interpretation.

3 Dark components: density profiles or DFs

3.1 Stellar DF with prescribed dark densities: the starting model

The observed stellar distribution constrains the total gravitational field. Within the smooth, stationary approximation used here, the field depends on the dark mass densities, not on the velocities of individual dark particles. We therefore need not specify or fit a DM or remnant DF to calculate the stellar observables. AGAMA explicitly supports density/potential components without corresponding DFs (Vasiliev, 2019, section 5.2).

For the halo, sample the target density directly:

$$\rho_{\text{DM}}(r) = \rho_{20} \frac{h_\gamma(r/r_s)T(r/r_t)}{h_\gamma(20 \text{ pc}/r_s)T(20 \text{ pc}/r_t)}, \quad h_\gamma(x) = x^{-\gamma}(1+x)^{\gamma-3}, \quad T(y) = e^{-y^2}. \quad (16)$$

This gives $\rho_{\text{DM}}(20 \text{ pc}) = \rho_{20}$ exactly. Compare a cored family ($\gamma = 0$) and an NFW-like cusp ($\gamma = 1$) separately, with sampling coordinates (ρ_{20}, r_s) and positive r_s, r_t . The outer cutoff r_t is a declared shape assumption, initially fixed within each run and varied in sensitivity tests. It is not forced to equal the stellar boundary. These are proposed reference profiles, not a claim that either is selected by the data.

A concentrated remnant example is the Plummer density,

$$\rho_{\text{rem}}(r) = \frac{3M_{\text{rem}}}{4\pi a_{\text{rem}}^3} \left(1 + \frac{r^2}{a_{\text{rem}}^2}\right)^{-5/2}, \quad M_{\text{rem}} \geq 0, \quad a_{\text{rem}} > 0. \quad (17)$$

Its normalization is a mass; the halo’s normalization is a *local spatial density*. The corresponding potential of either spherical component, with $\Phi_i(\infty) = 0$, is

$$\Phi_i(r) = -G \left[\frac{M_i(< r)}{r} + 4\pi \int_r^\infty s \rho_i(s) ds \right], \quad M_i(< r) = 4\pi \int_0^r s^2 \rho_i(s) ds. \quad (18)$$

Total halo mass is derived and may depend strongly on r_t and extrapolation beyond the stellar data. Sampling ρ_{20} makes its prior explicit; a change of coordinates alone does not remove prior sensitivity. A uniform prior in ρ_{20} and a uniform prior in $\log \rho_{20}$ are different models. We must specify bounds and test that choice. The no-DM case is a separate model with $\rho_{20} = 0$, not the lower endpoint of a logarithmic prior. Any model-averaged mass at zero also depends on model prior odds. Outer density choices can affect bound-orbit actions and escape energies, so their effect on ρ_{20} must still be tested.

For each likelihood trial, hold the stellar DF parameters and dark density parameters fixed. Compute Φ_{DM} and Φ_{rem} , then iterate equations (1)–(2): recalculate the stellar actions and frequency map, rebuild and normalize the contour DF, and update the stellar contribution to the total potential until convergence. This additional DF update is required by the prescription in Section 2.2. This retains stellar self-gravity; the cluster is not treated as a massless tracer. “Prescribed” means fixed within this equilibrium calculation, not fixed across the parameter search. The remnant component is optional, and the no-DM model sets $\rho_{\text{DM}} = 0$.

At fixed dark density profiles and stellar DF parameters, two different dark orbital distributions produce the same smooth potential and stellar likelihood. There is no direct information about those dark orbital distributions in this likelihood. A chosen dark DF can link orbital parameters to density and hence affect the stellar predictions indirectly; that link is an additional model assumption. We recommend the prescribed density formulation for the first compact-model fits.

3.2 Positive DFs for all components: a stronger alternative

The original proposal instead follows composite action-DF constructions (Pascale et al., 2018; Binney and Vasiliev, 2023). It replaces the prescribed dark profiles by

$$\rho_i(\mathbf{x}) = \int f_i[\mathbf{J}(\mathbf{x}, \mathbf{v}; \Phi_{\text{tot}})] d^3v, \quad i \in \{\text{DM}, \text{rem}\}, \quad (19)$$

and recomputes all component densities as the common potential converges. A non-negative DF is then required for every included extended component in that potential. This certifies more than a stellar equilibrium in prescribed dark gravity, although it still does not prove dynamical stability.

In this formulation the DM halo has its own mass, action scale, orbital prescription, core/cusp choice and outer truncation. A useful notation for a positive normalized component is

$$f_i(\mathbf{J}) = \frac{M_i}{(2\pi)^3 J_{0,i}^3 \mathcal{N}_i} F_i\left(\frac{\mathbf{J}}{J_{0,i}}; \boldsymbol{\lambda}_i\right), \quad \mathcal{N}_i = \int F_i(\mathbf{u}; \boldsymbol{\lambda}_i) d^3u, \quad (20)$$

where $F_i \geq 0$ and $0 < \mathcal{N}_i < \infty$. The dimensionless shape parameters $\boldsymbol{\lambda}_i$ must be stated for each model family. This is a normalization convention, *not yet a chosen halo or remnant functional form*. The action double-power-law and cored/truncated constructions of [Posti et al. \(2015\)](#); [Pascale et al. \(2018\)](#) provide candidates. An action-space cusp parameter is not automatically the spatial density slope in the final joint potential; the resulting $\rho_{\text{DM}}(r)$ must be measured.

In either formulation the remnant component should be centrally concentrated but independent of the extended halo. Its shape can initially be fixed while its mass and scale vary. The halo must not be treated as another stellar mass bin with stellar equipartition, nor forced to share the luminous truncation. Present-day equilibrium does not imply a particular adiabatic formation history. Tidal survival and stripping are separate dynamical constraints on acceptable present-day configurations.

3.3 Parameter budget and interpretation

Table 1: Physical coordinates for the primary spherical candidate, with one anisotropy transition and prescribed dark densities. The fixed exponent $n = 2$, halo slope choice and outer cutoff are declared assumptions.

Component	Free coordinates	Count
Stars	$M_\star, J_{0,\star}, \alpha, b_{\text{out}}, J_a$	5
Halo	ρ_{20}, r_s	2
Remnants	$M_{\text{rem}}, a_{\text{rem}}$	2
Stars + halo + remnants		9
Stars + halo only		7
Stars only		5

Light normalization, equivalently $\Upsilon_\star$ at fixed stellar mass, is additional: the first row of totals becomes ten fitted coordinates if it is the only extra parameter. Other observational nuisance parameters must also be counted. The two-transition anisotropy law adds two coordinates; releasing the halo cutoff adds one; a central BH adds $M_\bullet$ and requires revisiting the stellar regularity prescription. Distance, flattening, rotation and tracer extensions add their own parameters when introduced. No-DM and no-remnant comparisons remove each absent component’s irrelevant scale parameters. There is no universal parameter count for every extension.

In the full dark-DF alternative, masses and action scales may replace the halo/remnant density coordinates. Equal counts would not imply equal density families or priors. Avoiding dark DFs simplifies the construction and avoids unobserved orbital choices; it does not remove the degeneracy between stellar mass, remnants and DM.

3.4 When to construct dark DFs

The prescribed-density model guarantees neither that a dark profile has an equilibrium orbital realization nor that the complete system is stable. These are additional physical checks, distinct from fitting the stellar data. For selected fitted models we can attempt a spherical density-to-DF inversion in the *combined* potential and test non-negativity and density recovery. A successful isotropic inversion supplies one admissible DF. A negative result rules out that isotropic realization; it

does not by itself exclude all anisotropic realizations of the same density. Any consistency restriction imposed on the fit must state its orbital assumptions.

Live N-body experiments also need a velocity distribution for each sampled dark component. Different admissible DFs at the same initial density can respond differently to tides and heating. We can construct those DFs after the stellar fit and assess that dependence in the dynamical experiments, without adding unconstrained dark orbital parameters to the initial stellar likelihood. An explicit dark-DF inference model remains an alternative if we choose to impose that stronger equilibrium structure from the outset.

4 Observable predictions and fitting

4.1 Tracer selection is a prerequisite for DM inference

Partial energy equipartition can make different stellar-mass samples have different dispersions in the same potential. oMEGACat VI measures a radial variation in this effect (Häberle et al., 2025). HST, MUSE and Gaia magnitude cuts, crowding cuts and spatial footprints therefore need explicit comparison. Chemical populations are another possible distinction, but a metallicity split is not automatically a stellar-mass split. The published matched HST–MUSE comparison uses the same stars; we must establish which selection is represented by each profile used here rather than assume that all 89 bins have that property.

First audit each profile’s stellar-mass/luminosity range, membership criteria, error cuts, radial weights and rotation subtraction. Where catalogues allow, construct kinematics for overlapping mass and population selections, and fit those before combining unmatched profiles. If matching is impractical, require a minimal two-tracer recovery test in a common potential. The tracer DFs describe the selected populations; their number weights and light weights must follow the selection, while stellar mass is counted only once in Poisson’s equation. Adding another freely weighted copy of the whole stellar mass distribution would not perform this test.

Mocks should deliberately give the tracers different dispersions and spatial profiles, then measure the false ρ_{20} signal produced by forcing one tracer. This tests confusion with anisotropy, M/L gradients and remnants. A spherical DM result remains conditional until these controls establish its sensitivity to the actual catalogue selections.

4.2 Projection and sample weighting

For the same effective stellar tracer, let $j = \rho_\star/\Upsilon_\star$ and $\sigma_r^2 = p_r/\rho_\star$. At projected radius R ,

$$I(R) = 2 \int_R^\infty \frac{j(r)r \, dr}{\sqrt{r^2 - R^2}}, \quad (21)$$

$$I\sigma_{\text{LOS}}^2(R) = 2 \int_R^\infty \left(1 - \beta \frac{R^2}{r^2}\right) \frac{j\sigma_r^2 r \, dr}{\sqrt{r^2 - R^2}}, \quad (22)$$

$$I\sigma_{\text{PM,R}}^2(R) = 2 \int_R^\infty \left(1 - \beta + \beta \frac{R^2}{r^2}\right) \frac{j\sigma_r^2 r \, dr}{\sqrt{r^2 - R^2}}, \quad (23)$$

$$I\sigma_{\text{PM,T}}^2(R) = 2 \int_R^\infty (1 - \beta) \frac{j\sigma_r^2 r \, dr}{\sqrt{r^2 - R^2}}. \quad (24)$$

These expressions are for the spherical nonrotating baseline. The observed proper-motion dispersions follow by dividing velocities by $4.74047 D[\text{kpc}]$ to obtain mas yr^{-1} .

For a catalogue with projected number density $\Sigma_a(R)$ and radial selection $S_a(R)$, the model second moment in a complete annulus is

$$\langle v_k^2 \rangle_{a,b} = \frac{\int_{R_b^-}^{R_b^+} 2\pi R S_a(R) \Sigma_a(R) \langle v_k^2 \rangle_a(R) dR}{\int_{R_b^-}^{R_b^+} 2\pi R S_a(R) \Sigma_a(R) dR}. \quad (25)$$

The bin dispersion is the square root of the averaged variance, rather than the average of the local dispersion. Actual spatial footprints and existing bin kernels must be retained. Integrated light and a star-count-selected kinematic sample need not weight stellar populations identically. The single-tracer assumption is therefore explicit and must be checked for the HST, MUSE and Gaia samples.

In a future discrete-data likelihood, the DF is marginalized over unmeasured coordinates and convolved with velocity errors. Selection and contamination are part of that probability model. Conditional velocity likelihoods can be useful under specified selection assumptions (Vasiliev et al., 2026); their cancellation of spatial selection is not universal. Rotation also requires matching each data product’s treatment of mean streaming, not simply substituting raw second moments for dispersions.

4.3 Inference targets

The primary derived quantities are

$$\rho_{\text{DM},20} \equiv \rho_{\text{DM}}(20 \text{ pc}), \quad M_{\text{tot}}(< 20 \text{ pc}) = 4\pi \int_0^{20 \text{ pc}} (\rho_\star + \rho_{\text{rem}} + \rho_{\text{DM}}) r^2 dr + M_\bullet. \quad (26)$$

The stellar/DM decomposition can be uncertain even when total enclosed mass is well determined. Posterior uncertainties and sensitivity to halo shapes, stellar M/L assumptions and anisotropy must accompany any inferred density. The total outer halo mass is a different, extrapolation-sensitive quantity.

Where the errors are calibrated Gaussian measurements, a joint fit uses the appropriate covariance in $-2 \ln \mathcal{L} = \mathbf{r}^T C^{-1} \mathbf{r} + \ln \det C + \text{constant}$, where $\mathbf{r}$ is the data-minus-model residual vector. The existing mock programme’s adopted photometric weights are not a measured covariance: its score Q must remain a diagnostic score. Neither that score nor the lowest residual alone selects the physical model.

4.4 Rotation and flattening before final inference

The spherical fit is a recovery experiment and initial data comparison. The observed system has a central counter-rotating signal (Pechetti et al., 2024) and a flattened dispersion field (Häberle et al., 2025). A final DM interpretation requires an axisymmetric rotating comparison, or a quantitative demonstration that its omission does not affect the reported constraint at the stated precision.

For an axisymmetric model, decompose

$$f(J_r, J_z, J_\phi) = f_{\text{even}}(J_r, J_z, |J_\phi|) + f_{\text{odd}}(J_r, J_z, J_\phi), \quad f_{\text{odd}}(-J_\phi) = -f_{\text{odd}}(J_\phi). \quad (27)$$

The complete DF must remain non-negative; a sufficient bound is $|f_{\text{odd}}| \leq f_{\text{even}}$. The even part controls density and raw second moments, and the odd part sets streaming. Since

$$\sigma_\phi^2 = \langle v_\phi^2 \rangle - \langle v_\phi \rangle^2, \quad (28)$$

dispersions about the mean also depend on the odd part. A change in the sign of streaming may be needed for the core. Forward modelling must use the same spatial bins and local or global mean subtraction as the data. Neither a rotation correction nor spherical averaging establishes that flattening is harmless. The regularity conditions near $J_\phi = 0$ in an oblate potential need their own construction and tests (Binney, 2026); the spherical $L \rightarrow 0$ rule cannot simply be reused unchanged.

5 Model hierarchy and staged validation

Table 2: Roles of the proposed model families. The primary candidate and control must pass validation before a production inference run.

Family	Role
Regularized exponential action DF with a transition	Primary spherical candidate
Generalized lowered-isothermal/LIMEPY	Required independent stellar control
Linear-action exponential precursor	Analytic checks and literature reproduction
Three-component flexible action mixture	Approximation capacity and misspecification tests
Explicit DFs for stars and dark components	Stronger equilibrium and live N-body validation

5.1 The independent lowered-isothermal control

Use the generalized lowered-isothermal family (Gieles and Zocchi, 2015, 2018), which has already been applied to ω Cen (Zocchi et al., 2017, 2019):

$$f_{\text{LI}}(E, L) = A \exp \left[-\frac{L^2}{2r_a^2 s^2} \right] E_\gamma \left(g_t, \frac{E_t - E}{s^2} \right), \quad E < E_t, \quad f_{\text{LI}} = 0 \quad (E \geq E_t), \quad (29)$$

where $E_\gamma(0, x) = e^x$ and $E_\gamma(g_t, x) = e^x \gamma(g_t, x) / \Gamma(g_t)$ for $g_t > 0$. Here $\gamma(g_t, x)$ is the lower incomplete gamma function; g_t is a truncation-shape parameter, distinct from the contour slope $g(J_r, L)$. The Michie–King case is $g_t = 1$. Require $A, s, r_a > 0$, $g_t \geq 0$, and $E_t < 0$ for a finite boundary when $\Phi(\infty) = 0$.

The energy is computed in the combined star+DM+remnant potential and stellar self-gravity is retained. An isolated LIMEPY solution with a halo added subsequently would not implement this control. Embedded energy-DF constructions also provide precedents (Amorisco and Evans, 2011, 2012). This family is distinct from replacing E by $E + L^2/(2r_a^2)$ inside a lowered exponential.

Its cluster motivation is strong: a nearly isotropic central region, radial bias at intermediate radii and a finite energy boundary. Its standard anisotropy law returns towards isotropy near the edge and cannot freely represent tangential outskirts. Moreover, our spherical potential excludes the nonspherical Galactic tide, so its cutoff is an effective boundary, not an exact tidal escape surface. The action envelope is also a phenomenological outer taper; it does not solve tidal evolution. We choose the action family as primary for its orbital flexibility and route to axisymmetric modelling, and require the energy family as a different physical control. Agreement in ρ_{20} after matched selection and common dark-profile/normalization priors is more informative than agreement among close variants of one DF.

5.2 Checks required before production fitting

We propose the following sequence.

1. **Construction and regularity.** Check contour endpoints and numerical mass normalization against the harmonic limit; test $b = 0$ against an ergodic construction. Compare approximate and exact frequency ratios. Measure $f_{J_r}/f_L \rightarrow 2$ and the cancellation in equation (5), including velocity distributions near $v_t = 0$ and the joint action origin. Double action, velocity, contour and potential resolutions. Check stellar density and force convergence from independent starting potentials. Positivity alone is not a pass.
2. **Anisotropy and recovery.** Establish which $\beta(r)$ profiles are actually generated by each transition law. First use matched-family mocks to check optimization, then independent stars-only and stars+DM mocks, including concentrated remnants, halo-shape mismatch and M/L gradients. Distinguish fixed-potential approximation tests from mass recovery with an unknown potential. Use several starting parameter sets.

3. **Independent control and noise.** Fit the generalized lowered-isothermal model to the same mocks and selected data. Test recovery bias, scatter and false DM signals, with explicit ρ_{20} priors and outer-cutoff sensitivity. A small noise ensemble is a pilot, not a coverage certification. Retain family differences in the inferred uncertainty.
4. **Observed tracer selection.** Match mass/luminosity samples or perform the two-tracer test in Section 4.1. Do this before assigning cross-catalogue residuals to DM or adding dark orbital parameters.
5. **Geometry and dynamical admissibility.** Assess rotation and flattening as in Section 4.4. Construct dark DFs for selected posterior density profiles in the combined potential and test live stability and tidal survival separately. Preserve the observed surviving cluster as a constraint on evolution experiments.

Central-BH experiments must allow the stellar inner structure to respond. The central slope–anisotropy restrictions of [An and Evans \(2006\)](#), including $\gamma \geq \beta + 1/2$ for the Keplerian limit, provide useful necessary checks; they do not replace a DF construction. Stability and survival in a Galactic tide require separate time-dependent experiments. A non-negative steady DF alone certifies neither.

6 Relationship to the implemented model

The existing AGAMA note, `agama_df_models.tex`, describes the implemented stellar double-power-law family. Its mass-recovery extension has three six-parameter stellar shapes, independent stellar mass/light weights, and positive dark *density* templates, for 27 fitted coordinates. The compact proposal replaces the stellar mixture by one regularized exponential action DF with an anisotropy transition. Its recommended starting version retains dark density templates; the all-component DF version is an optional extension. The existing code thus already uses the proposed separation between stellar phase space and dark gravity, but the compact stellar family is not yet implemented or validated.

The earlier capacity test established that a mixture could approximate its independent finite-boundary mocks to stringent accuracy. It did not establish that the real data require three stellar terms. Conversely, 27 parameters alone does not prove overfitting. This proposal asks whether a more deliberate family improves identification of the physical masses without unacceptable bias from restricted density, anisotropy or M/L freedom.

Before a new production run, the contour and frequency-map implementation, prior ranges, tracer selections and the independent lowered-isothermal control must pass the tests above. The next deliverable is therefore a compact model comparison and validation report. The existing batch remains a separate flexibility experiment, with its results and frozen inputs preserved.

A Reproducibility of this note

The three illustrations are generated by `bin/plot_compact_df_proposal.py`. They use analytic functions and a component diagram only; no simulated or observed mass-recovery results enter them. PNGs live in `plots/`, and the plotted arrays and provenance in `results/plot_data/compact_df_proposal.json`. The companion literature review is `docs/DF_MODEL_LITERATURE_REVIEW.md`. The bibliography is the ADS-verified `docs/df_model_literature.bib`. Build products and page renders belong in `results/diagnostics/compact_df_writeup/`; the final PDF lives beside this LaTeX source in `docs/`.

B The analytic precursor, retained for diagnostics

The unregularized exponential-action family of [Pascale et al. \(2019\)](#) is

$$f_{\text{lin}}(J_r, L) = C_{\text{lin}} \exp \left[- \left(\frac{J_r + \eta L}{J_{0,\text{lin}}} \right)^\alpha \right], \quad (30)$$

$$C_{\text{lin}} = \frac{M_\star \eta^2 \alpha}{(2\pi J_{0,\text{lin}})^3 \Gamma(3/\alpha)}, \quad \eta > 0. \quad (31)$$

The exact mass follows from $2 \int dJ_r \int dL e^{-[(J_r + \eta L)/J_{0,\text{lin}}]^\alpha} = J_{0,\text{lin}}^3 \Gamma(3/\alpha) / (\eta^2 \alpha)$. This useful positive family reproduces literature examples and checks integration routines, but its derivative ratio is $1/\eta$ everywhere. It is not the reference production candidate. Only $\eta = 1/2$ obeys the finite-centre regularity limit; even then global isotropy is not automatic. For the harmonic $b = 0$ contour check, the scale conversion is $J_{0,\star} = 2J_{0,\text{lin}}$, not equality of the two scale parameters.

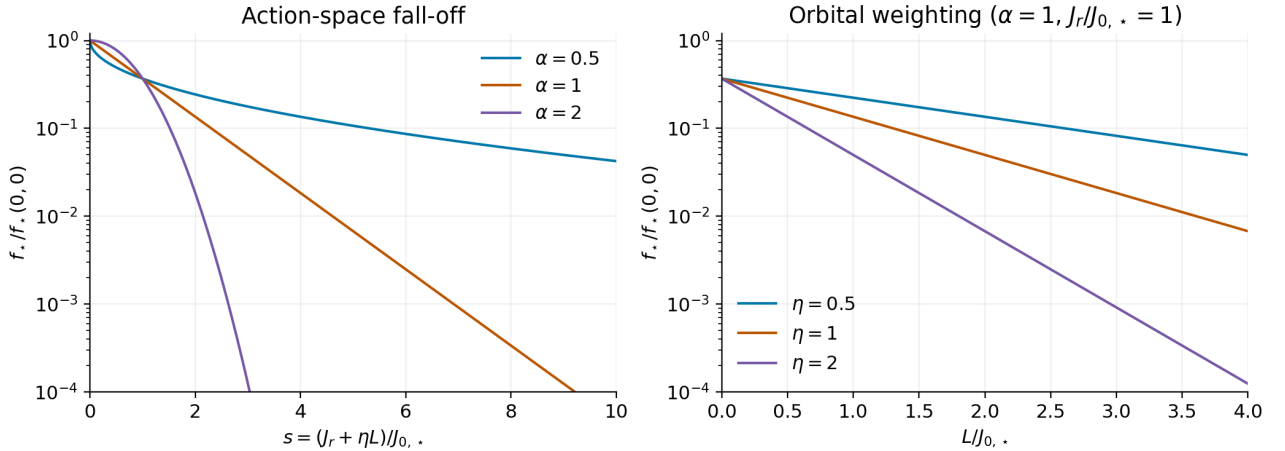


Figure 3: Analytic slices of the diagnostic precursor, equation (30), divided by their own zero-action values. The different α and η curves illustrate envelope and orbital weighting; their mass-normalization constants differ. These are not density profiles, $\beta(r)$ predictions or the proposed regularized DF.

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